EuqVid

A Multimodal Approach to Gender Bias Detection in Video Datasets

Research Structure, Component Specifications and End-to-End Workflow
Group J26-DS-343 · CoEAI · SLIIT · IT4010 Research Project
Supervisor: Dr. Prasanna S. Haddela · Parent methodology: EuqAud (Jayawardena et al., IEEE Access 2026)

> **Status**
> *Working plan for group agreement and supervisor review. Section 9 lists what remains open. Claims attributed to prior work were recalled or verified during drafting and must be checked against the source before they enter the TAF or the thesis.*

1. The research, in one paragraph

Video datasets used to train AI systems systematically under-represent one gender — not only in how many people appear, but in screen time, framing prominence, speaking duration and the language written for them. There is currently no way to measure this before training. It surfaces afterwards as model performance disparity, when it is expensive to fix, and models do not merely inherit dataset bias but amplify it (Zhao et al., EMNLP 2017). EuqVid produces a single interpretable score from -10 to +10, computed directly from raw visual, temporal, textual and acoustic features, as a frozen closed-form equation rather than a retrained model — so it is auditable and comparable across datasets.

What EuqVid measures: *within-content* representational disparity. Given subjects of both genders present in the same material, is one systematically afforded greater prominence? Zero means parity, not the naturally occurring distribution.

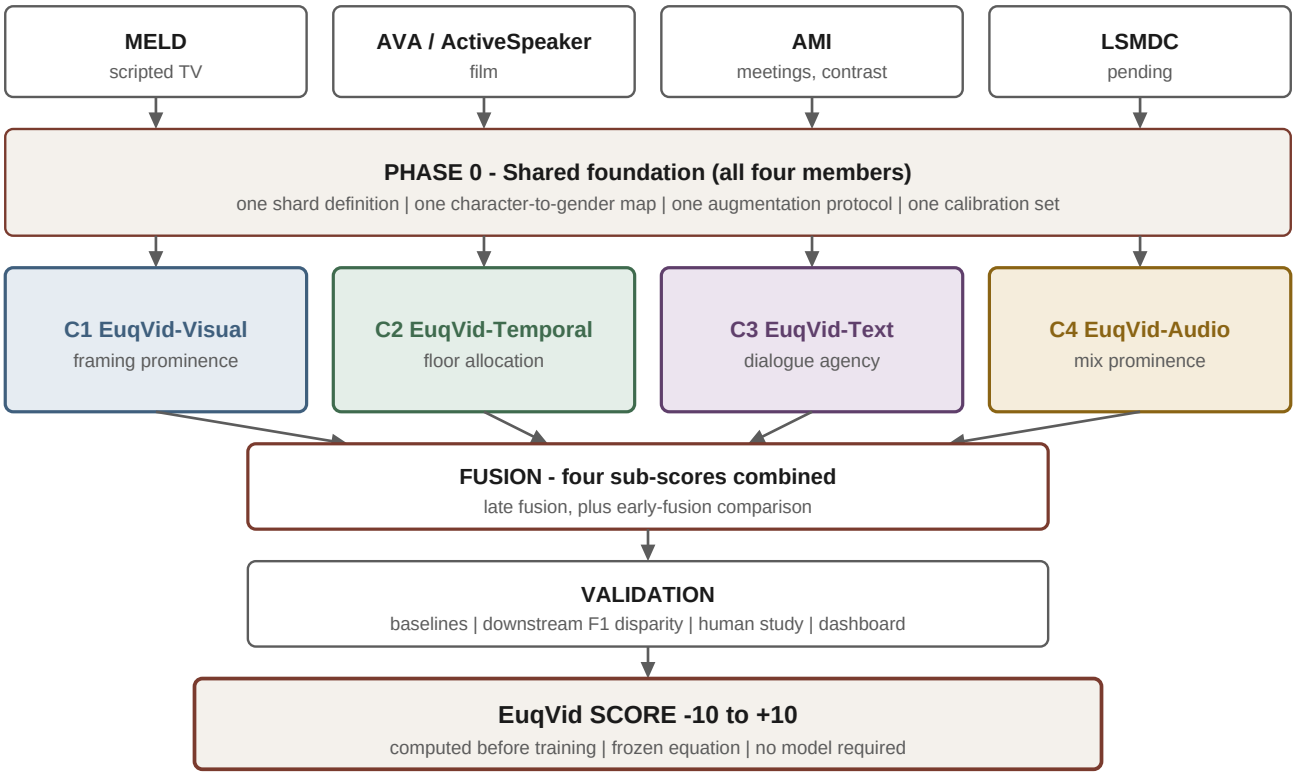

Figure 1 — Overall research flow, from corpora to the final score.

2. The four components

	Full title	Short name	Measures
C1	EuqVid-Visual: A Visual Prominence-Based Approach to Gender Bias Detection in Video Datasets	Visual-EuqVid	What the frame gives each gender
C2	EuqVid-Temporal: A Temporal Analysis of Gender Representation Dynamics in Video Datasets	Temporal-EuqVid	What the timeline gives each gender
C3	EuqVid-Text: A Dialogue-Based Approach to Gender Bias Detection in Video Datasets	Textual-EuqVid	What the script gives each gender
C4	EuqVid-Audio: Adapting Acoustic Gender Bias Detection to Multi-Speaker Video Audio	Audio-EuqVid	How each gender's voice is treated in the sound
	EuqVid: A Multimodal Approach to Gender Bias Detection in Video Datasets (team thesis)	EuqVid	The fused score

> **The boundary rule — agree this before anyone writes code**
>
> ***Visual*** *owns anything measured in* ***the frame****.* ***Temporal*** *owns anything measured in* ***time****.* ***Text*** *owns* ***words****.* ***Audio*** *owns anything measured in* ***sound****.*
>
> *So speaking duration is Temporal, because it is seconds. Voice loudness is Audio, because it is decibels. This is the boundary most likely to drift during implementation.*

2.1 The pipeline every component follows

All four components run the same twelve-step pipeline. Only the extraction stage differs.

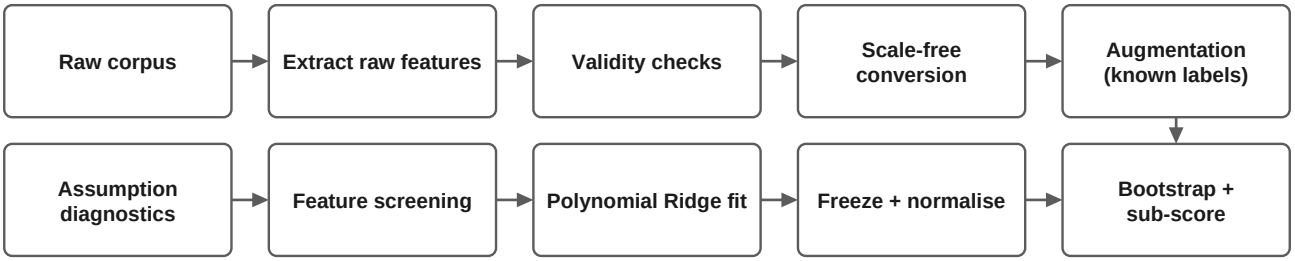

Figure 2 — The common component pipeline, identical for C1 to C4.

C1 — EuqVid-Visual

EuqVid-Visual: A Visual Prominence-Based Approach to Gender Bias Detection in Video Datasets

	Statement
Problem	Video datasets under-represent one gender not only in how many people appear, but in how they are framed — how large, how centrally, how often alone in shot. Existing dataset documentation records none of this, so it cannot be corrected before training.
Gap	GD-IQ and the USC Annenberg reports count screen time and speaking characters. Mazieres, Menezes and Roth (2021) computed a female face ratio across 3,776 films. All of them count *presence*. None measures *prominence* — size, centrality, shot scale, solo framing — and none produces a dataset-level score a builder can act on.
Methodology	Extract per-gender framing statistics from face annotations: ground-truth boxes and entity IDs on AVA-ActiveSpeaker, a detector elsewhere. Convert to scale-free shares. Generate known-bias shards with the shared augmentation protocol. Fit polynomial Ridge, freeze, normalise, bootstrap.
Novelty	The first framing-based gender bias score for video datasets — measuring how each gender is composed within the frame rather than merely whether they are present, as a frozen closed-form equation with a declared applicability domain.

Extracted features

- **screen_time_share** — face-frames of this gender divided by all face-frames
- **mean_relative_face_area** — average face size as a fraction of the frame area
- **mean_centre_proximity** — average closeness to the middle of the frame
- **closeup_share** — proportion of this gender's frames that are close-ups
- **solo_frame_share** — how often they appear as the only face in frame
- **identity_share** — distinct characters of this gender divided by all characters

Tasks, beginning to end

1. Acquire face annotations. On AVA-ActiveSpeaker use the shipped entity_box and entity_id fields, which removes detector error entirely for that corpus. On MELD and AMI run MTCNN or an equivalent detector.
2. Annotate gender per entity_id / character using the shared group map. Report the excluded share for ambiguous identities rather than silently dropping them.
3. Assign clips to shards under the group shard contract.
4. Compute raw per-frame statistics: face area, centre distance, faces present, identity counts.
5. **Detector fairness check.** Measure detection rates by gender and, where annotation permits, by skin tone and pose. Quantify how much detector disparity propagates into the score. This is a viva question you will be asked; have the number.
6. Manual validation: hand-annotate a sample and compare against the automatic pipeline.
7. Aggregate to shard-by-gender level and convert to scale-free shares and ratios.
8. Apply the shared augmentation protocol to create shards at known bias levels.
9. Run assumption diagnostics: Durbin-Watson, Breusch-Pagan, VIF.
10. Rank predictors by Pearson and Spearman correlation with the label.
11. Screen features inside CV folds, then grid-search polynomial degree and Ridge alpha.
12. Freeze the equation in raw feature units, normalise on the shared calibration set, derive tiers, run the 1,000-iteration bootstrap, and hand the sub-score to the fusion stage.

C2 — EuqVid-Temporal

EuqVid-Temporal: A Temporal Analysis of Gender Representation Dynamics in Video Datasets

	Statement
Problem	Disparity in who holds the floor, for how long, and at what points in the narrative is documented by media analysts but has never been turned into a computable dataset property.
Gap	GD-IQ measures how much women speak. Garnerin, Rossato and Besacier (2019) found women under-represented in speech turns and among prominent speakers in broadcast corpora, and showed it degrades ASR performance — but on broadcast news, and not as a dataset score. Nobody measures how the floor is *allocated*: turn structure, interruption, or whether speech is spread through the story or bunched into one scene.
Methodology	Derive speaking intervals from ground truth where available: AVA-ActiveSpeaker's per-frame speaking labels, MELD's CSV timings, AMI's manual transcripts. No lip-activity detector or VAD is required on these corpora. Compute turn-level statistics, convert to scale-free form, augment, fit, freeze.
Novelty	Turn-taking structure and temporal dispersion as dataset bias signals, together with the presence-voice gap — visible but silent — which no single-modality measure can detect.

Extracted features

- **speaking_time_share** — speech seconds of this gender divided by all speech seconds
- **turn_count_share** — how many times they are given the floor
- **mean_turn_length** — how long they hold it, normalised by the shard mean
- **interruption_asymmetry** — how often they are cut off, minus how often they cut others off
- **speech_dispersion** — how evenly their speech is spread across the timeline. A character who speaks throughout is woven into the story; one who speaks heavily in a single scene is not
- **presence_voice_gap** — screen-time share minus speaking-time share. On screen but silent

Optional / conditional:

- **mean_shot_duration** — average length of shots featuring this gender — film only
- **cut_share** — how often the edit cuts to this gender — film only

> **Acquire speaking intervals. AVA-ActiveSpeaker ships SPEAKING_AND_AUDIBLE labels per face track; MELD ships StartTime and EndTime per utterance; AMI ships word-level manual transcripts.**
>
> *Attach gender to each speaking interval using the shared group map.*

Tasks, beginning to end

1. Collinearity warning
2. Speaking time, turn count and mean turn length are algebraically linked: total time equals turns multiplied by mean length. VIF will be severe and you should expect to keep two of the three. That is why **speech_dispersion** and **presence_voice_gap** are on the list — they keep the component above four features after screening.

   The two optional features are censored on MELD, whose clips average 3.59 seconds, because a shot cannot be longer than the clip containing it. Report them as a film-only sub-analysis, never inside the frozen equation.

   presence_voice_gap requires C1's screen-time share as an input, so extraction must be shared.

C3 — EuqVid-Text

EuqVid-Text: A Dialogue-Based Approach to Gender Bias Detection in Video Datasets

	Statement
Problem	The transcripts that ship inside video datasets are training inputs, not context. MELD is an extension of the text-only EmotionLines corpus, so its emotion labels were annotated from dialogue in the first place. Bias in that text propagates into any multimodal model trained on the dataset.
Gap	NLP has measured gendered film dialogue extensively — Sap et al. (EMNLP 2017) found female characters written with systematically less power and agency; Agarwal et al. (NAACL 2015) automated the Bechdel test. But all of this work analyses *complete screenplays*, which are pre-production artefacts never distributed with video datasets. Nobody has measured the transcripts that actually ship inside video corpora, and nobody has produced a dataset-level text bias score.
Methodology	Read utterance-level text with speaker labels directly from annotation files — no video processing is required. Map speakers to gender with the shared map. Dependency-parse for subject-verb pairs and look verbs up in the connotation-frame lexicon for agency and power. Aggregate, convert to scale-free form, augment, fit, freeze.
Novelty	The first dialogue-level gender disparity score computed on the text that ships inside video datasets — the text a multimodal model actually ingests — rather than on screenplays.

Extracted features

- **dialogue_volume_share** — words spoken by this gender divided by all words
- **mean_utterance_length** — average words per line, normalised by the shard mean
- **agency_score_mean** — mean agency value of verbs for which this gender is the syntactic subject
- **power_score_mean** — mean power value over the same subject-verb pairs
- **same_gender_dialogue_share** — proportion of their lines in adjacent turns with the same gender
- **mention_ratio** — times named by other characters divided by times they speak
- **sentiment_valence_mean** — mean sentiment polarity — ships free with MELD

> **Read the annotation files. MELD's train, dev and test CSVs carry Utterance, Speaker, Emotion, Sentiment, StartTime and EndTime. AMI requires parsing the manual annotation XML. No video is opened at any point in this component.**
>
> *Build the character-to-gender mapping and publish it as an appendix. Report the share of utterances excluded for unknown or ambiguous speakers.*

Tasks, beginning to end

1. Corpus coverage
2. **AVA ships no transcripts**, so this component cannot score the film corpus as things stand. Three resolutions, and one must be chosen in Phase 0: (a) align film subtitles to AVA-ActiveSpeaker's per-frame speaking labels to recover speaker-attributed dialogue — the only fix that closes the gap completely; (b) use LSMDC descriptions instead of dialogue, which changes the construct from what characters say to how they are described; (c) compute the fused score in a documented reduced form where text is unavailable, and validate that it tracks the full form on MELD and AMI.

   Note also that **utterance_count_share was removed** because it is the same quantity as C2's turn_count_share, and counting it twice would double-weight it in the fusion.

C4 — EuqVid-Audio

EuqVid-Audio: Adapting Acoustic Gender Bias Detection to Multi-Speaker Video Audio

	Statement
Problem	EuqAud measures acoustic gender bias in clean, single-speaker speech corpora. Produced narrative audio is a different object: voices overlap, and a mix engineer decides how far each voice sits above the music and effects. That mixing decision is an authored prominence choice, directly analogous to shot scale in the visual domain, and no metric measures it.
Gap	EuqAud's corpora — Common Voice, LibriSpeech, TED-LIUM, AMI — are recorded in quiet conditions, so a speech-to-background ratio cannot even be defined in them. A search for research on gender and film sound mixing returns nothing academic. Meanwhile EuqAud's own baseline analysis found pitch, amplitude and energy differences to be weak standalone indicators, with an effect size below 0.18 against 0.39 to 0.42 for the count and voice-activity ratios.
Methodology	Separate voice from background with a source-separation model. Attribute speech segments to gender using the shared map and, on AVA, the face-linked active-speaker labels. Compute mix-prominence features alongside EuqAud's transferred standard-deviation features. Convert to scale-free form, augment, fit, freeze.
Novelty	Mix prominence as a gender bias channel — a signal structurally impossible in EuqAud's data — with EuqAud's transferred features serving as the declared baseline rather than the contribution.

Extracted features

- **mean_voice_loudness** — how loud their voice is relative to the shard's overall dialogue level
- **speech_to_background_ratio** — how far their voice sits above the music and effects bed
- **dynamic_range** — the spread between their loudest and quietest delivery
- **f0_std** — pitch variation. Standard deviation, never the mean — mean pitch would only detect that a woman is speaking
- **energy_std** — energy variation. Transferred from EuqAud
- **amplitude_std** — amplitude variation. Transferred from EuqAud

> **Extract the audio track and confirm sample rate and channel layout per corpus.**
>
> *Run source separation to split dialogue from music and effects. Validate the separation on a hand-checked sample before trusting it at scale.*

Tasks, beginning to end

1. Baseline versus contribution
2. The first three features are the contribution; the last three are EuqAud's verbatim and are the **baseline**. State that split explicitly in the thesis. Dr. Haddela supervised EuqAud, and AMI is already inside EuqAud's training base, so the panel will know precisely what is incremental.

   Because no prior work suggests the mix-prominence effect exists, **design the component so that a null result is still a reportable finding.**

7. The end-to-end workflow

Phase 0 — Shared foundation (all four members, first three weeks)

1. Settle the open decisions in Section 9: the text corpus question, the shard definition, the labelling protocol and the calibration set.
2. Freeze the feature schema in writing, signed by all four members.
3. Build **one** character-to-gender map used by all four components. Without it the sub-scores describe different populations and cannot be fused.
4. Write down the boundary rule: frame is Visual, time is Temporal, words are Text, sound is Audio.
5. Run a coverage pilot on ten clips per corpus. For every feature ask three questions: can I compute it here, does it mean the same thing on every corpus, and does it vary at all.
6. Write the drop rule **before** seeing any results — a feature is dropped if it fails coverage, if VIF exceeds 10, or if variance is near zero. Dropping features after seeing which perform well is researcher degrees of freedom and an examiner will name it.
7. Freeze one augmentation protocol. Four private label definitions cannot be fused into one score.
8. Agree the identity-level partition firewall in writing, so validation never runs on the shards the equations were fitted on.

Phase 1 — Extraction (all four in parallel)

9. Each member extracts their channel's features, per shard, per gender.
10. Convert everything to shares and ratios. No raw seconds or raw counts survive this step: a raw magnitude grows with dataset size and would score a large balanced corpus as more biased than a small skewed one.
11. Merge into one versioned feature file.

Phase 2 — Build the four sub-metrics (all four in parallel)

12. Apply the shared augmentation to create shards at known bias levels.
13. Screen features inside cross-validation folds, never on the full set beforehand.
14. Fit polynomial Ridge regression with degree capped at 2.
15. Freeze each equation in raw feature units, folding scaling constants into the published coefficients.
16. Normalise to -10 to +10 on the shared calibration set, so +6 denotes the same disparity in all four.
17. Derive Neutral, Moderate and Strong thresholds from stated disparity levels rather than inheriting EuqAud's 2 and 6.
18. Bootstrap 1,000 times, resampling at shard and identity level — never at row level, which produces fictitiously narrow intervals on synthetically expanded data.

Phase 3 — Fusion

19. Collect the four sub-scores per shard.
20. Fit the fusion over the four scores, including pairwise interactions.
21. **Also** fit one single equation over all features directly, as the comparison. Late fusion cannot see cross-channel patterns such as a character who is on screen constantly but silent. Reporting both is a result either way.
22. Freeze EuqVid.

Phase 4 — Validation

23. Compare EuqVid against three baselines — face-count ratio, screen-time ratio, speaking-time ratio — using Kruskal-Wallis effect size. Compare against EuqAud's reported figure rather than pre-committing to beating it.
24. Train downstream classifiers on subsets spanning the disparity range. MELD ships emotion labels; AVA-ActiveSpeaker ships active-speaker labels.
25. Measure F1 and accuracy disparity per subset and correlate against the EuqVid tiers.
26. Run the mitigation experiment: rebalance the worst subsets by the score, retrain, and re-measure the harm. This tests whether the metric is *actionable* rather than merely descriptive — a step EuqAud never took.

27. Run the human perception study, 15 to 20 participants, on clips spanning the disparity range. Start the ethics application in week one; approval latency is external.
28. Cross-check against AMI's DOME dominance annotations for free independent human judgement.
29. Build the Streamlit audit dashboard with SHAP per-feature attributions.

Phase 5 — Write up

30. Four individual theses and one group thesis.
31. Report the feature reduction as a finding, not an apology. Beginning with twenty-seven candidate features and retaining a documented subset is a stronger sentence than quietly using the survivors.

8. The final outcome

One number per dataset, computed before any model is trained, from a frozen equation that fits on a single page and requires no classifier at any stage.

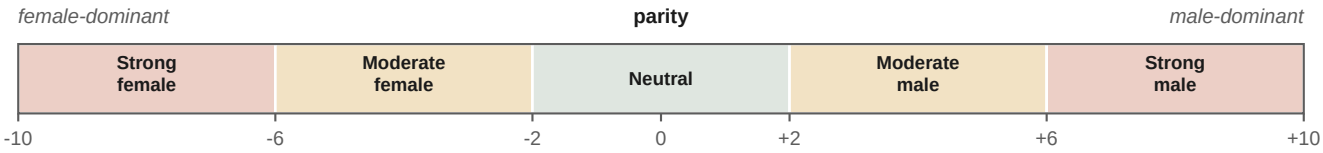

Figure 3 — The EuqVid score, direction and severity tiers.

Deliverables

	Deliverable
D1	Four frozen closed-form sub-equations in raw feature units, plus the fused EuqVid equation, all with published coefficient tables.
D2	Per-shard scores on the -10 to +10 scale with Neutral, Moderate and Strong tier assignments, and the derivation of the thresholds.
D3	Ranked feature tables per component showing which signals carry bias information and which do not, with Pearson and Spearman correlations.
D4	Bootstrap stability tables: mean, standard deviation and 95 per cent confidence interval at score and tier level, with the resampling unit stated.
D5	Baseline comparison with effect sizes, and the downstream F1 disparity correlation.
D6	The mitigation result — whether rebalancing a dataset by the score measurably reduces downstream disparity.
D7	Human perception study results and agreement with the tiers.
D8	A Streamlit audit dashboard with SHAP attributions, and the shared assets: the character-to-gender map, excluded-speaker shares, and per-feature coverage rates.
D9	A limitations register: what each component cannot measure, and on which corpora it cannot run.

The claim the research can support, if the results hold

Gender representational disparity in a video dataset is measurable before training, across four modalities, on a single interpretable scale, using frozen equations that require no retraining and can be audited line by line — and acting on that score measurably reduces downstream model performance disparity.

9. Still to settle

	Item	Why it matters
9.1	**Which text source for C3?** Dialogue (MELD, AMI), LSMDC descriptions, or subtitle alignment onto AVA.	AVA ships no transcripts. This decides whether the fused score can be computed on film at all, and whether the component title remains accurate.
9.2	**Does the shard contract survive MELD?** Identity partitioning requires that no character spans two shards, but Friends has six leads appearing in nearly every episode.	Two hundred identity-disjoint shards cannot be built from six recurring identities. Either shard by episode and document the leakage, or accept fewer shards and revisit the coefficient budget.
9.3	**Who owns Phases 3 and 4?** All four members are on a modality, so fusion, validation, the perception study and the dashboard belong to nobody.	Name one of the four as integration lead and place Phases 3 and 4 in the group thesis under that name. Unowned work is unassessed work.

	Item	Why it matters
9.4	**Is the TAF revisable?** It says "using visual and temporal features," which no longer describes a four-modality project, and it repeats two claims known to be wrong.	Open since the first restructure document. Confirm before rewriting anything.

Corrections to carry into the TAF regardless of any other decision

1. *The TAF states twice that EuqAud "explicitly identified video as future work." It does not. Section V covers demographic attributes, multi-speaker audio, far-field and noisy conditions, the scarcity of female-biased corpora, and synthetic-label realism. Section VI says "other demographic attributes and* ***multimodal data****."*

2. *"8+ datasets" against five named and three secured. Fix the number.*

3. *"8+ independent datasets" is not independent without the identity-level firewall.*

4. *"No collection of personal, sensitive, or identifiable data" while processing faces and inferring gender. Faces are biometric data. The position is defensible on public corpora under existing licences, but it must be reworded rather than asserted.*

Working plan for Group J26-DS-343. Feature definitions, tasks and the workflow are proposals for group agreement, not settled fact. No results in this document are our own measurements; the only measured figure referenced is the MELD main-cast speaking-time ratio of 1.094 male to female, computed by the group.